Project: Smart Health Checkup Recommendation Dashboard

Purpose of conducting the POC: To assess the vendor's capability in developing an AI system that recommends health check programs in accordance with medical standards (Medical Logic & AI), has a flexible architecture that supports international standards (FHIR), and ensures Security and Privacy in compliance with PDPA / HIPAA standards.

Part 1: Topics the Vendor Must Present (POC Deliverables)

For the POC presentation, the Vendor must prepare and present the following 4 main topics:

1. Process Flow Architecture (system architecture and data flow)

● Explain the workflow of the Hybrid Recommendation Engine (Rule-based combined with AI).

● Show the structure for retrieving data from the Hospital Information System (HIS) through Middleware such as the Model Context Protocol (MCP) Server.

● It must be demonstrated that MCP Tools are restricted under Least Privilege and operate in read-only mode for data retrieval before sending data to AI.

2. Mockup Screens (user interface design - minimum 3 screens)

● Screen 1: Individual Summary & Recommendation

Must display the recommended health check program, together with the AI's "Logical Reasoning (Explainable AI)" and "Confidence Score" clearly.

● Screen 2: Trend & Timeline

Must display graphs showing health data trends over the past 3–5 years, together with a timeline of significant events (e.g., abnormal values or disease diagnosis).

● Screen 3: Access Monitoring Dashboard

A screen for system administrators (Admin/DPO) focusing on alerts for abnormal usage behavior (Anomaly Detection).

3. Architecture & Data Standard Strategy

● Present a Pluggable AI architecture that can demonstrably switch AI models (e.g., from GPT-4 to Claude 3.5 or Local Llama) through a configuration file or setup screen without modifying the source code.

● Explain the Terminology Mapping strategy from hospital codes (Local Code) to the HL7 FHIR R4 standard using LOINC (for Lab) and SNOMED CT (for Condition / Procedure).

4. Security, Privacy & Audit Strategy

● Explain the approach for using the SMART on FHIR standard to define the scope of data access.

● Present the process for performing Data Masking or pseudo-anonymization to remove Personally Identifiable Information (PII) before sending data into external AI processes.

● Show the format for maintaining comprehensive Audit Logging that records all 5 dimensions (Who, When, Where, What, Why), including logging of the AI Prompt / Context, with logs that must be immutable.

Part 2: POC Evaluation Criteria

The evaluation of the POC presentation will be based on the following topics, with a total score of 100%.

Evaluation Area	Details Looked for by the Evaluation Committee (Key Focus)	Score Weight
1. Recommendation Logic and System Architecture/Flow	• Clarity of the hybrid mechanism (Rule-based + AI) in selecting the primary program and additional examination items. • Understanding of the use of the MCP Server as an intermediary for retrieving data in read-only mode and enforcing Least Privilege access control.	35%

2. Data Standards and Architectural Flexibility	• Practical feasibility of the Pluggable AI architecture that allows model switching through a config file or setup screen without modifying the code. • Reliability of the terminology mapping strategy to the HL7 FHIR R4 standard (LOINC, SNOMED CT).	25%
3. UX/UI Design and Explainability	• The Individual Summary screen clearly presents Reasoning (Explainable AI) and the Confidence Score in a logical and easily understandable manner. • Effective presentation of historical health data trends and a practical anomaly alert window that can be used in real scenarios.	20%
4. Security, Privacy, and Auditing Capabilities	• Robustness of the Data Masking / pseudo-anonymization process before sending data to AI. • Design of a 5-dimension Audit Log system, including immutable storage of Prompt / Context in accordance with PDPA standards. • Application of SMART on FHIR for permission management.	20%
	Total	**100%**